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Implementation and Control of Cooling Subsystem for an ME3 Photon Detector Enclosure

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This report documents the development and testing of the ME3 X-ray photon detector enclosure cooling subsystem, following on the enclosure and cooling architecture developed by Cheng [1]. Three deliverables are presented. First, the previously proposed cascaded PID control architecture was re-evaluated against the actual thermal operating constraints of the ME3 detector, which was found to be a 40°C safety threshold rather than a precise setpoint temperature. This meant that a temperature banded open loop PWM scheme that modulates the pump duty cycle in discrete steps depending on measured temperature was enough for the cooling system. Secondly, the coolant loop was physically assembled, integrating a 12V four-wire PWM pump, water reservoir, U-shaped copper pipes and a 180W thermoelectric heat exchanger into a sealed closed loop. Thirdly, the control system is coded and prototyped on a breadboard, consisting of a Raspberry Pi Pico 2W microcontroller, a MAX31850K thermocouple amplifier with 1-Wire interface, a Omron D6F MEMS airflow sensor, a BSS138-based bidirectional shifter providing 3.3V to 5V translation for the pump's PWM and tachometer signals and an IIC character Liquid Crystal Display (LCD) with an audible buzzer and visual warning indicators.

Control firmware was implemented in MicroPython and organized around two independent subsystems running concurrently on the Pico. For the cooling subsystem, the MAX31850K reads detector temperature, applies temperature banded PWM to modulate the pump duty cycle, and reports any thermocouple fault conditions. The drying subsystem reads the Omron D6F MEMS airflow sensor monitoring air passing through a desiccant and raises warnings if airflow falls below the threshold required to maintain enclosure humidity below the dewpoint of coldest expected coolant surface.

Future pending work for the subsequent development cycle is discussed, which include PCB fabrication from the prototype breadboard, on detector integration testing, full integration of the dry-air subsystem under closed loop humidity feedback and chiller capacity assessment under the full multi-detector heat load.

Keywords—ME3 X-ray photon detector cooling, thermal management, PWM control, PID control, Raspberry Pi Pico 2W, MicroPython, Thermocouple Instrumentation, Level-Shifted Logic Interface.

I. PROJECT BACKGROUND

A. Introduction

The ME3 X-ray photon detector enclosure project is an effort within UC Berkeley's Department of Mechanical Engineering, with the goal of housing arrays of Detection Technology's ME3 multi-energy X-ray detectors [2] [3].

These enclosures are to be installed on a rotating and translating stage in Etcheverry Hall's FLOW LAB, which will be used for multiphase fluid analysis via X-ray radiography. Each enclosure must be able to support and shield the detectors mechanically, manage their thermal heat dissipation, and prevent condensation damage during cooling system operation.

The central challenge is to manage the thermal environment of the detectors. The ME3 X-ray photon detectors have a published operating temperature range of 0°C to 40°C [2]. Each detector dissipates approximately 28W of power via heat under normal operating conditions [3], giving a six-detector enclosure a heat load of 168W. The detectors themselves have a built-in mechanism to protect against overheating, which is that reversible power down begins above 55°C at the Cadmium Telluride ($CdTe$) crystal or 75°C at the electronic board [2]. In extreme cases, a non-recoverable hard shutdown may occur when temperatures exceed 60°C at the crystal. Therefore, active cooling must be implemented to refrain shutdown of operations at such high temperatures. Without active cooling, thermal characterization results from Cheng [1] extrapolates the FPGA-region temperature to 69°C within ten minutes of operation, which is well past the safe operating temperature range.

This report documents the continuation of the project across the 2025/26 academic year. The scope of the project was to take the previously proposed cooling architecture from an on-paper design to a working physical and electronical prototype. Three contributions are presented: a re-evaluation of the proposed control architecture, the physical assembly of coolant and dry-air subsystems, and the design and implementation of the sensors in an associated control system.

B. Previous Work

The project has been worked on across multiple academic years. Earlier contributions by *Kolby Gameros* and *Dylan Anacleto-Black* established the initial enclosure concept and identified a high-level cooling strategy. In the most recent academic year, efforts from *Jonathon Cheng* delivered the bulk of mechanical design and the analytical groundwork that this report builds upon.

Cheng's work included the manufacturing of two aluminum 6061 enclosures, the geometric design of U-shaped copper coolant pipes, and the use of finite element analysis to validate the enclosure structure under gravitational loading on the rotational stage. The coolant pathway was specified as 3/8 inch copper pipe in a U-shape running the length of the detector array. The pipes enter the enclosure on the FPGA side, mounted with thermal glue to maximize heat-transfer surface contact, making a loop and exiting on the ADC side. Cheng specifically avoids using any fluid connectors in the enclosure, with all joints occurring on the outside of the enclosure with 1/2 inch NPT cable connectors to eliminate any possibility of leaking over the detectors.

Sizing calculations were also completed by Cheng, who found the pump head to be approximately 3.5m based on the Hazen-Williams approximation for the pipe geometry that was manufactured. A 12V centrifugal pump rated for 800l/h and 4m head was selected on this basis. The chiller capacity was sized to 180W given the 168W heat load.

Cheng also developed a dry-air subsystem strategy, due to the coolant pipes operating near the dew point of unconditioned ambient air. The reason for this was to prevent condensation on cold pipes from dripping onto the detectors below it. The strategy he employed was to push ambient air through a desiccant chamber and into the enclosure, to maintain interior humidity at below 40% relative humidity. At this humidity level, the dew point is 15°C, which is well below the coldest expected temperature of the coolant pipes.

Finally, Cheng proposes a cascaded PID control architecture for the coolant system of the enclosure. The outer loop regulates detector temperature by setting a target coolant temperature, while the inner loop regulates coolant temperature by modulating the pump RPM and chiller fan speed via PWM. This architecture was not implemented, and outstanding tasks such as coolant-loop assembly, electronics integration, control loop implementation and bench validation were all explicitly identified as outstanding tasks.

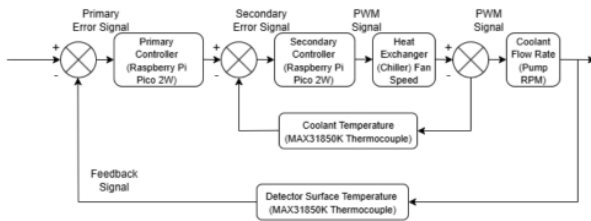


Fig. 1 The proposed cascade feedback coolant control system diagram from [1] with an inner and outer loop.

C. Project Specific Objectives

For this academic year, the work conducted is grouped under four main objectives:

- **Control system evaluation and redesign:** The cascaded PID proposal was assessed against the system

requirements. Since it is a one-sided safety threshold of 40°C and not a precise setpoint, the complexity of the PID system is determined to be unnecessary. Instead, a simpler temperature banded PWM modulation system is found to be sufficient for our use case. More on this decision is expanded upon in the upcoming sections.

- **Physical assembly of coolant and dry air subsystems:** The previously manufactured U-pipes, water pump, reservoir and thermoelectric heat exchanger were physically integrated into a sealed closed water loop. Each fitting was threaded, sealed and leak-tested before electrical integration. The dry air path is equipped with airflow sensing to allow for fault detection, with section IV detailing the physical assembly.
- **Electronics design and prototyping:** A breadboard prototype of the full sensor and switching circuit is constructed around the Raspberry Pi Pico 2W microcontroller. The breadboard incorporates a MAX31850K thermocouple, an Omron D6F MEMS air-flow sensor, a BSS138 bidirectional level shifter, an IIC character LCD, a buzzer and warning LED lights. The full list of hardware choices is shown in Section III, with a complete schematic done in KiCAD and presented in Section VIII.
- **Control firmware implementation:** Code is developed in MicroPython to support two subsystems running in parallel for the same microcontroller. In essence, a stepped PWM control of the coolant pump based on detector temperature, and a monitoring of dry air flow rate with audio and visual warnings. Implementation details are shown in Section VI.

Having achieved these four objectives, the validated breadboard circuit is transferred onto a KiCAD schematic with annotated layouts to support handover to the next development cycle. This paper then discusses PCB fabrication, on-detector integration testing and migration to closed loop PID control as future targets in Section IX.

II. ELECTRICAL SYSTEM ARCHITECTURE

This section presents the system in terms of electrical organization, and details such as part selection and pin wiring are discussed in subsequent sections. The purpose of this section is to establish how the system is partitioned into voltage domains, how signals cross between them and how power is distributed across the subsystems.

A. Logic Level Management

The system operates across three different voltage domains. The Raspberry Pi Pico 2W and all sensors run on 3.3V logic. The pump's PWM control input and tachometer output operate at 5V logic, consistent with the four-wire fan and pump standard given by the component manufacturer. The pump motor and the thermoelectric chiller draw their power from independent 12V rails.

Each domain exists for varying reasons. The 3.3V level is fixed by the Pico's specification, the 5V level is fixed by the pump's signal interface, and the 12V level is due to the motor and Peltier's power requirement. All three of them cannot be collapsed into a single rail without breaking any one of the interfaces.

The boundary that requires active management is the 3.3V $\leftrightarrow$ 5V signal interface. A 3.3V output from the Pico cannot be registered as a logic on the 5V input of the pump. On the other hand, a 5V tachometer pulse driven into the Pico's GPIO pin exceeds its absolute maximum voltage rating and can damage it. Therefore, some level of translation is required between these two logic levels.

This role is carried out by the BSS138 bidirectional level shifter. The shifter occupies a column of bidirectional channels between the Pico (LV side) and the pump's signal lines (V side). One channel carries the PWM command from the Pico to the pump, which is then translated from a 3.3V logic upwards to a 5V logic. The opposite process occurs from 5V to 3.3V from the pump's tachometer pulses to the Pico. The MOFSET topology of the BSS138 shifter is bidirectional, hence a single channel can handle both signal directions on the same wire without any directional ambiguity.

Finally, the 12V domain is deliberately isolated from the logic, without interacting with the Pico, shifter or any sensors. The pump speed is varied by changing the duty cycle of its 5V PWM input, not by varying its 12V motor power. This keeps high current paths separated from sensitive analog transducers and eliminates the need for switching transistors or relays. Failure modes in the 12V domain cannot directly damage the logic side of electronics, and the two domains meet only at the common ground.

B. Power Distribution Strategy

The system uses three independent power sources, summarized as shown below:

- **12V, 2A power supply for pump**

A dedicated 12V DC supply supplies power to the motor for the centrifugal pump. This supply is rated according to the pump's continuous current draw and is adequate for 800l/h and 4m head performance for the pump.

- **12V, 15A power supply for thermoelectric chiller**

A second, independently rated 12V DC power supply is used for the thermoelectric chiller. This supply is sized for an operating current of 15A to give a total power rating of 180W. The chiller is operated as an always-on load whenever the system is powered on and is not switched or modulated by the Pico in the current set up.

The current disparity between the pump and chiller is the reason why a shared power supply is not used. A combined power supply would have required a single 17A unit that is expensive and harder to procure. Combining them would also risk the high current draw of the chiller disrupting the PWM

modulated current draw of the pump. Instead, the decision to isolate both power supplies allows for two independent power sources that do not interfere with one another.

- **Pico-derived rails**

The Pico is powered via Micro USB from a host computer for the duration of our bench development. The MicroUSB provides a nominal 5V at the VBUS pin and supplies the current required by the Pico itself and the sensors. The Pico's onboard switching generator then produces the 3.3V logic rail which is connected to the 3V3 output pin. This rail powers all of the logic-side components such as the thermocouple, flow sensor, LCD module, LV side of bidirectional shifter, buzzers and LEDs. The 5V logic rail is powered by VBUS and powers the HV side of the bidirectional shifter. No external 5V supply is required for the breadboard prototype in this case, although there is room for improvement during the operation of the enclosure and its subsystems.

- **Common ground**

All three power supplies share a common ground reference to ensure uniformity of logic between all systems. The 12V supplies' negative pins, the pump and chiller return, the Pico's ground pin and the ground of all sensors are connected to a common ground.

This is extremely essential as the system relies on signals defined as voltages between conductor and ground. Without a single shared reference, the logic domain would drift relative to each other and cause erratic sensor readings and unreliable PWM behavior. This common ground setup forces high-current and low-level returns to meet at a single point, isolating their voltage drops from one another.

C. Block Diagram

Referring to Figure 2, each block represents a functional unit rather than an individual component. Arrows dictate the direction of power or signal flow. Solid arrows denote power while dashed arrows denote control or sensing signals.

The block diagram is divided into four functional groupings:

- **Power domain (Blue)**

The three independent power supplies for the pump, chiller and Pico with a common ground tie point.

- **Logic domain (White)**

The Pico acts as a central node, receiving sensor input and emitting control outputs.

- **Sensor domain (Orange)**

This is made up of the thermocouple and air flow sensor to feed temperature and flow rate data to the Pico.

- **Actuation domain (Green)**

The diagram illustrates the power and signal connections for a Raspberry Pi 2W system. A legend indicates that solid arrows represent power connections and dashed arrows represent signal connections. A diamond symbol represents a common ground.

Power Connections (Solid Arrows):

- 5V Supply:** Provides power to the Raspberry Pi 2W (V5V pin, 3.3V), the 12V PWM Pump (12V), the Peltier Thermoelectric Chiller (12V), the LCD Display (3.3V), and the Buzzer + LEDs (3.3V).
- 3.3V Supply:** Provides power to the Raspberry Pi 2W (V3V3 pin, 3.3V), the BSS138 Bidirectional Level Shifter (3V pin, 3.3V), and the Buzzer + LEDs (3.3V).
- 12V PWM Pump:** Provides power to the Peltier Thermoelectric Chiller.
- Motor Power:** Provides power to the 12V PWM Pump.

Signal Connections (Dashed Arrows):

- MAX3180K Thermocouple:** Provides a temperature reading to the Raspberry Pi 2W (Temp. Reading).
- D6F Air Flow Sensor:** Provides an airflow reading to the Raspberry Pi 2W (Airflow Reading).
- Raspberry Pi 2W:** Provides a PWM 3.3V to 5V signal to the BSS138 Bidirectional Level Shifter.
- BSS138 Bidirectional Level Shifter:** Provides a tachometer 5V to 3.3V signal to the BSS138 Bidirectional Level Shifter.
- BSS138 Bidirectional Level Shifter:** Provides a visual and audio warning signal to the Buzzer + LEDs.
- BSS138 Bidirectional Level Shifter:** Provides a display temperature and warnings signal to the LCD Display.

Common Ground: All blocks share a common ground.

D. Microcontroller

The **role** of the microcontroller is to act as a central control node to read sensor inputs from the thermocouple amplifier and air flow sensor. It then computes pump PWM duty cycle and warning conditions, to be sent to the display, buzzer and indicator LEDs.

The Raspberry Pi Pico 2W is built around the RP2350 dual-core ARM Cortex-M33 processor running up to 150MHz, with 520 KB of on-chip SRAM and 4MB of onboard flash memory. It has 26 multi-function GPIO pins, with support for features such as PWM, SPI, IIC and 1-Wire via its subsystem. It has an onboard buck-boost switching regulator which generates the 3.3V logic rail, allowing the board to be powered by a 5V power supply at VBUS. This rail powers all of the logic-side components in the system. MicroPython is supported, allowing the control firmware to be written in Python and updated by file copy.

E. Transducers

1) Thermocouple

The **role** of the thermocouple is to provide temperature feedback for the cooling control loop, and it is mounted at the FPGA-region of the detector to read its temperature.

A K-type thermocouple is used in combination with the Adafruit MAX31850K Thermocouple Amplifier Breakout. The breakout mounts the Maxim Integrated MAX31850K cold-junction thermocouple-to-digital converter on a small PCB. Some other components include a 3.3V LDO regulator, capacitors, screw terminal block for the thermocouple wires and accessible in headers.

The MAX31850K accepts input from K-type thermocouple and reports temperature with a 14-bit resolution (temperature steps of 0.25°C) over the temperature range of -270°C to 1372°C. The chip integrates cold-junction compensation internally hence there is no need for an external reference junction.

The breakout is powered by the Pico's 3.3V rail and is connected via a single line to the DATA pin. There is also a 4.7kΩ pull up resistor to the 3.3V as specified by Maxim's 1-Wire timing requirements. The K-type thermocouple has a glass-braid fiberglass insulation and a 2mm diameter stainless steel sheathed tip that is 2cm long, with the total wire length up to 1m. The fiberglass insulation allows the probe to be used up to 500°C, and it is mounted against the detector's FPGA region with the thermal glue.

2) Flow Sensor

The **role** of the airflow sensor is to monitor airflow through the desiccant chamber of the dry-air subsystem. This ensures that dry air is constantly flowing so that interior humidity is low enough to prevent formation of condensation on cold coolant pipes.

The airflow sensor is an Omron D6F-V air velocity sensor from the D6F MEMS flow sensor family. It uses a

microfabricated thermal flow element, where a heated MEMS structure is suspended between two on-die temperature sensors. The temperature difference between upstream and downstream sensors is found to be proportional to mass air velocity. The sensor contains no moving parts and is rated for $\pm 5\%$ of full-scale accuracy within its operating range of 0 – 3ms^{-1} .

F. Bidirectional Level Shifter

The **role** of the bidirectional level shifter is to translate logic-level signals between the Pico's 3.3V GPIO domain and the pump's 5V signal domain. The shifter translates the Pico's outgoing PWM signal upward to 5V to drive the pump's input and then translates the incoming tachometer pulses downward to 3.3V so it can be safely read by the Pico's GPIO.

The shifter is an Adafruit 4-channel Bi-directional Logic Level Converter, built around a BSS138N-channel MOSFET. Each of the bidirectional channels combines a BSS138 transistor with two 10kΩ resistors to translate logic signals in either direction over the same wire. The shifter can operate between 1.8V and 10V on either side, encompassing the 3.3V ↔ 5V domains in use for our use case.

For our current implementation, only two out of four channels are in use, one for outgoing PWM signal to pump and the other for incoming tachometer signal from pump. The low voltage (LV) side is powered from the Pico's 3.3V rail, while the high voltage (HV) side is powered from the Pico's 5V VBUS pin.

G. Interface

1) LCD Display

The LCD display is a GreekPi 128 x 64-pixel monochrome OLED module, built around the SSD1306 driver IC. It communicates with Pico over IIC on a dedicated SDA/SCL pair, with power being drawn from the 3.3V rail.

The OLED display supports rendering for both multi-line text and graphics, providing flexibility for displaying continuously changing sensor data. It also offers better viewing angle and contrast compared to other displays which may help with debugging in the future from arbitrary angles.

2) Buzzer

The buzzer used is an active buzzer which contains an internal oscillator that produces an audible tone when DC power is applied. To turn on the buzzer, a single command from the GPIO pin is needed with no PWM tone-generation logic required in the firmware. The buzzer operates directly on 3.3V from the Pico's GPIO pins.

3) Indicator LEDs

Two LEDs provide visual warning indicators, with one for over-temperature conditions and one for low-airflow conditions. Each LED is driven from a Pico GPIO pin through a 1kΩ resistor to limit the current. The LED forward current is limited to around 2mA which makes it visible at indoor

lighting levels. Different colors are used for LEDs to allow for operators to instantly identify the system's warning states.

IV. MECHANICAL ASSEMBLY

This section documents the physical mechanical assembly of both the cooling loop and dry air subsystem. The work in this section builds on upon the previously designed components – manufactured aluminum enclosures, copper U-pipes and CAD designs for air handling modules by Cheng in [1]. These are integrated with the pump, chiller, desiccant chamber and 3D printed air detection module into two sealed and functional fluid systems

A. Cooling Loop Subsystem Assembly

The coolant loop circulates water from a reservoir, through the pump, into the inlet of the thermoelectric chiller for heat rejection. The water then flows through the U-shaped copper pipes and absorbs heat from the detector array inside the enclosure, before returning to the reservoir to complete a full loop. This flow sequence follows the architecture established by Cheng [1] as shown in Figure 3 below.

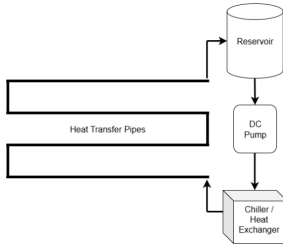


Fig. 3 Flow sequence described by Cheng in Figure 20 of the report cited in [1].

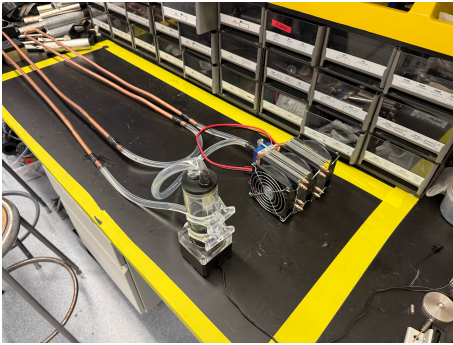


Fig. 4 Mechanical assembly of the pump, chiller and coolant pipes with sealed fittings.

Tubing and Adapters

Clear PVC soft tubing with 3/8" internal diameter for all flexible fluid segments. There are two main reasons why soft tubing is chosen over copper as the fluid tubing. It accommodates the relative positioning of bench components without having to be too precise on their locations, and it provides visual confirmation of fluid travel during leak testing. The copper U-pipes inside the enclosure remain in place as rigid heat transfer segments, while soft tubing carries

water externally between the copper pipes, pump, chiller and reservoir.

The FREEZEMOD pump has three G1/4 threaded ports which act as water inlet and outlet ports. To mate the pump's G1/4 threads with the 3/8" soft tubing, a brass G1/4 to 3/8" adapter was installed at each port. The adapter threads into the G1/4 port on one end and provides a barbed shank on the other end for the soft tubing to slide over. The same adapter is used at the chiller's water ports. This piece is standard solution for interfacing PC-style water cooling components with soft tubing and provides a neat solution for transitioning between components and soft tubing.

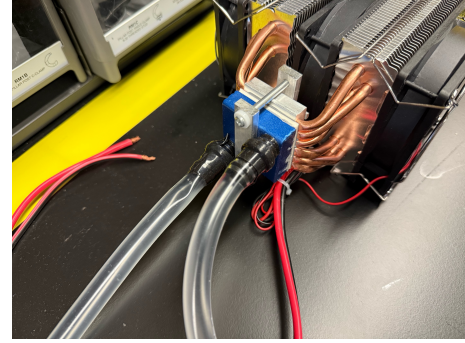


Fig. 5 The adapter to mate G1/4 ports to the 3/8" soft tubing.

Sealing Strategy

A two-layer sealing strategy is applied at every fluid joint to ensure the watertightness of our entire system. All threaded joints were first wrapped with PTFE (Teflon) thread sealant tape before threading. The PTFE tape fills in the helical gap between male and female threads, providing a primary seal against water leaks. The soft tubing to barb joints were then secured with waterproof tape wrapped externally over tube and adapter shank, providing a secondary seal in case the primary thread seal wears off over time under thermal cycling.

This double layer approach added negligible cost and assembly time and proved to be effective during bench leak testing. The tube was filled with water and operated for over 30 minutes without any observable leaks at the fittings. Reservoir water level was also monitored before and after the test, showing no measurable drop, hence confirming a sealed system.

B. Dry-Air Subsystem Assembly

The dry-air subsystem prevents condensation on the cold coolant pipes by maintaining the humidity of the enclosure below the dew point of the coldest expected pipe surface. This strategy, dew-point analysis and target humidity of 40% EH were established by Cheng in [1].

The subsystem operates as a single-pass airflow. Ambient air is drawn through a desiccant chamber by an air pump, which dries the air which is then delivered into the enclosure interior. The displaced air then circulates through the enclosure before exiting via the air detection module.

Continuous airflow is essential to make sure that fresh dry air constantly replacing the static air that has absorbed moisture in the enclosure.

The subsystem is made up of two main components, which are called the **air generation module** and **air detection module** respectively.

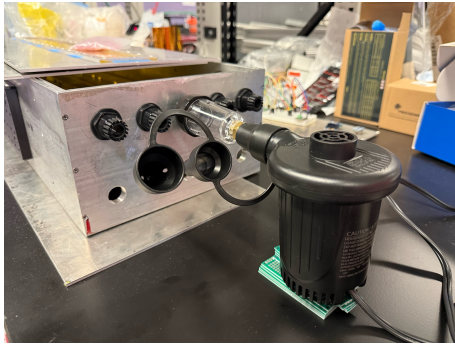


Fig. 6 Air generation module with air pump fitted directly to the desiccant chamber to produce airflow.

Air Generation Module

The air generation module consists of the Dr.meter HT-166 electric air pump [12] fitted to a dedicated desiccant chamber. The HT-166 is a 110V, 130W electric pump that is marketed for inflating air mattresses. The pump operates from a wall outlet, is separate from the system's DC power architecture and is always kept running when the system is powered.

A deliberate design decision was made to choose the electric air pump over the originally considered PC-style cooling fan. As mentioned in [1], the enclosure is designed to maintain a slight positive internal pressure, while the desiccant chamber introduces additional flow restrictions due to the silica gel medium. A PC fan can generate substantial airflow in open conditions; however, it would produce negligible pressure differential and stall against the back pressure of the sealed enclosure and desiccant chamber. The HT-166 has a 0.7 psi pressure capability and can comfortably overcome both restrictions, delivering airflow effectively into the enclosure. The pump is also rated for 460l/min flow capacity which exceeds the dry-air delivery requirement for this enclosure volume.

The desiccant chamber was originally meant to be a two-component design, with a 3D printed funnel directing fan air into a desiccant chamber. The solution that was built removes the need for a funnel, with the chamber having two NPT1/2" threaded fittings, one as an inlet and the other as an outlet. Silica gel is loaded into the desiccant chamber, and the pump's exhaust nozzle is connected to the chamber's NPT fitting. The chamber's outlet thread screws directly into one of the pre-machined NPT1/2" port holes on the enclosure side wall, as shown in Figure 6.

Air Detection Module

The air detection module houses the Omron D6F-V03A1 air flow sensor inside a 3D printed structure designed to mount directly onto the enclosure side wall. The module incorporates an integrated NPT1/2" threaded boss that screws into the second of the pre-machined enclosure side-wall ports, which is positioned directly under the port of the air generation module as shown in Figure 7.



Fig. 7 Enclosure sidewall with threaded fittings for air generation module and air detection module.

The internal geometry of the module uses a Venturi effect to enhance sensitivity, with the flow channel narrowing in cross section towards the position of the sensor. This increases the local air velocity past the sensor element. This is important as the air velocity inside the enclosure is expected to have a low bulk flow rate, and therefore having this effect increases the velocity at the sensor by a cross-sectional ratio. This amplifies the result and allows the firmware to easily distinguish if flow is present in the enclosure.

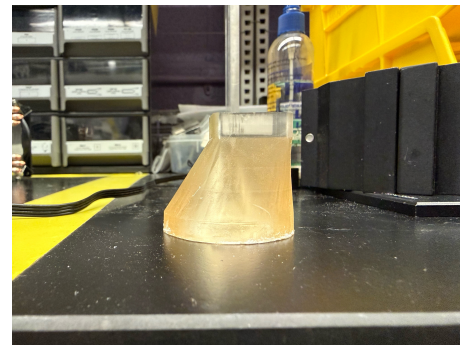


Fig. 8 The air detection module produced with 3D resin printing designed by Cheng in [1], with the air flow sensor fit tightly into it.

C. Integration with Control Electronics

The mechanical assembly is interfaced with the breadboard-based electronics in three main places.

(a) Thermocouple Mounting

The K-type thermocouple is mounted on the FPGA-region of the detector board where heat is dispersed most intensely. Thermal glue is applied between thermocouple tip and mounting surface to maximize

thermal coupling. The thermocouple wires then exit the enclosure through dedicated wire ports and terminate at the screw block of the MAX31850K breakout.

(b) Pump signal wiring

The four wires from pump exit from bench setup and terminate at the breadboard. It is housed separate from the enclosure, with a dedicated housing unit for all electronics and subsystem components.

(c) Power and sensor cabling

The chiller's 12V power cable connects directly to its dedicated 15A supply, bypassing the breadboard since the chiller remains turned on for the entire time. The air pump power is connected to a wall outlet but on the same circuit as the rest of the bench setup. The rest of the sensors are all powered by the Pico's rails as mentioned before.

During bench testing, the breadboard is positioned on the workbench adjacent to the assembled coolant loop and air-handling modules. Final integration onto the rotating stage will require either a permanent mounting of the breadboard or its PCB successor to be designed, which is addressed as a future task.

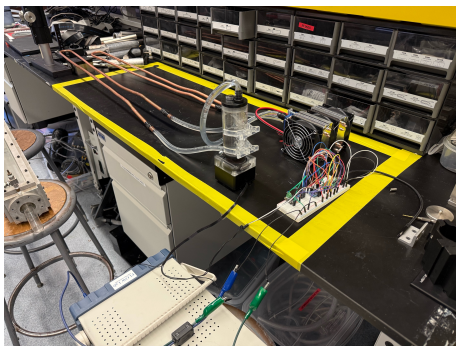


Fig. 9 The full cooling subsystem setup with electronics and physical assembly.

V. CONTROL SYSTEM DESIGN

This section presents the control architecture for the cooling subsystem. We briefly discuss the initially proposed cascaded PID control loop in Figure 1. It then moves onto the development of the temperature banded open-loop PWM scheme that was eventually implemented. The control of the dry-air subsystem is also discussed as a threshold-based fault monitoring system rather than a continuous control loop.

The existing hardware fully supports the migration of the controller to a closed-loop PID architecture, with steps required discussed in Section VIII.

A. Re-evaluation of Cascaded PID Architecture

Cheng proposes a cascaded PID architecture for coolant temperature control, with an outer loop regulating detector temperature and inner loop regulating coolant temperature through modulating pump speed and chiller fan duty cycle.

Cascaded control is the ideal choice when the thermal system must hold a precise setpoint despite disturbances. It is used in cases such as precision spectroscopy, where the CdTe crystal must be held at a fixed temperature to improve energy resolution.

In our case, the operating requirement for this system is rated for a 0 – 40°C range with over-temperature protection beginning at 55°C. The dry air subsystem also imposes that the coolant temperature must be near 15°C to remain above the dew-point margin. Therefore, the constraint for this system is a one-sided safety threshold, not a precise setpoint. There is no big difference whether the detector is held at 22°C or 25°C, as long as they do not exceed the operating temperature range.

Cascaded PID control requires two thermocouples and tuning six gain values which depend on empirical bench experiments. It also modulates the actuators continuously, increasing wear and tear on the pump and chiller.

B. Temperature-Banded Open-Loop PWM

The implemented control logic modulates the coolant pump's PWM duty cycle as a discrete function of measured detector temperature. The operating law behind it is a piecewise constant function – the temperature axis divided into different bands with each band mapping into a fixed duty cycle. The pump is the only controlled actuator, as the thermoelectric chiller is operated as always on.

Table II shows the temperature band and the corresponding pump duty cycles:

TABLE II. TEMPERATURE TO DUTY CYCLE MAPPING

Temperature Range	Pump Duty Cycle
$T < 20^{\circ}\text{C}$	0% (Pump off)
$20^{\circ}\text{C} \leq T < 25^{\circ}\text{C}$	25%
$25^{\circ}\text{C} \leq T < 30^{\circ}\text{C}$	50%
$30^{\circ}\text{C} \leq T < 35^{\circ}\text{C}$	75%
$T \geq 35^{\circ}\text{C}$	100%

The control law operates without a precise setpoint or error signals. At each control update every 0.5s, the firmware reads the current detector temperature from the MAX31850K. It then identifies which band the temperature falls into and sends the corresponding PWM duty cycle to the pump via the level shifter.

The threshold values were chosen to span the detectors 0 – 40°C operating range, with 5°C of headroom below the upper limit and the same margin above the dew point temperature. This ensures that the system never goes too low or high in temperature, but threshold values can be further refined once more testing data is available.

An over-temperature warning, consisting of the audible buzzer and temperature indicator LED fires at the warning temperature of 38°C, several degrees above the 100% duty cycle. This is meant to serve as a warning that the temperature is still rising despite the cooling system being saturated at maximum duty cycle.

There are three main benefits of such an architecture. Firstly, unlike a relay loop, the banded scheme produces a gradual response with the pump duty cycle stepping up accordingly as the temperature rises, which reduces overshoot compared to on/off control. Secondly, the band boundaries and duty cycles are chosen directly by design choice, requiring none of the system identification or gain tuning that PID demands. Finally, the discreteness of the output means that even if there are slow temperature fluctuations, there will be no pump response, hence eliminating any possibility of oscillatory instability.

The principal tradeoff against PID is the issue of fidelity. The banded scheme is unable to hold the detector at a specific temperature, as it can only keep it within a band's worth of tolerance ($\pm 5^\circ\text{C}$). For the ME3's safety constraints, this is sufficient for the use case.

C. Dry-Air Subsystem

The dry air subsystem does not include a formal control loop. The air pump runs continuously whenever the system is powered on, and there is no actuator that is under firmware control. The role of the firmware in this subsystem is for fault monitoring only, with the air velocity sensor sampling the air flow rate constantly. If the airflow drops below a certain threshold value indicating blockage or air pump failure, an audible buzzer and indicator LED is activated to alert the operators.

The reason for this simplification is because there is no benefit for constant modulation of the airflow rate. The dry-air system must be able to deliver sufficient dry air into the enclosure to keep it at a certain humidity, and the amount of dry air does not matter in this instance.

VI. SOFTWARE IMPLEMENTATION

The control firmware for the cooling and the dry-air subsystems was implemented in MicroPython [14] and runs on the Raspberry Pi Pico 2W. The firmware is structured as a single loop that runs at iterations of 500ms. This is short enough given the system's slow thermal dynamics, as any fluctuations in temperatures will take seconds to happen. The choice of IDE used for coding is Thonny in our case.

The firmware is presented below in functional blocks, with each subsection describing one component of the system. The code is then shown with comments explaining the design choices made. The goal is that any future students referring to this may find it easy to understand the justifications made and make any changes if needed.

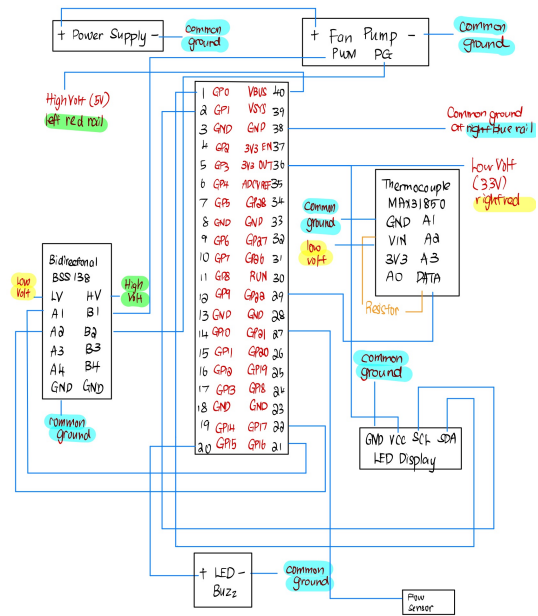


Fig. 10 Figure showing the GPIO connections for breadboard connections.

The complete physical wiring of the breadboard prototype is shown in the figure below. Every GPIO referenced in the firmware corresponds to a labelled connection in the figure.

A. Imports and Driver Installation

Before any work can be done, the libraries on which the firmware depends on must be installed. These libraries are the **machine**, **time** and **onewire** module for 1-Wire bus communication. The third-party **ssd1306** driver found at [15] has to be installed by copying it to the Pico while connected to USD with Thonny's IDE package manager.

```
import machine
import time
import onewire
from machine import Pin, ADC, PWM, I2C
import ssd1306
```

B. Configuration Constants

All tuneable parameters must now be defined as named constants before the main code. This includes the temperature band thresholds, the over-temperature warning threshold of 38°C, the airflow warning threshold, and calibration constants for airflow sensor. Future students may edit these constants as needed if there is any change during operation of the system.

```
# Control loop timing
LOOP_PERIOD_MS = 500 # Main loop update interval (ms)

# Temperature band thresholds (deg C) per Section V.B Table II
TEMP_BAND_OFF = 20.0 # Below this: pump off
TEMP_BAND_LOW = 25.0 # Above this: 25% duty
TEMP_BAND_MID = 30.0 # Above this: 50% duty
TEMP_BAND_HIGH = 35.0 # Above this: 75% duty
```



```

TEMP_WARNING          = 38.0      # Above 35: 100% duty
temp_warning          # Above this: trigger

# Flow threshold (m/s) - to be tuned from bench
# measurements
FLOW_WARNING_MIN      = 0.3      # Below this: trigger
flow_warning          # flow warning

# D6F-V03A1 calibration (voltage at sensor output to
# air velocity)
# voltage divider needed.
FLOW_V_AT_ZERO        = 0.5      # 0.5V at 0 m/s
FLOW_V_AT_FULL        = 2.5      # 2.5V at 3 m/s
FLOW_RANGE_MS         = 3.0      # Full-scale range for
D6F-V03A1
ADC_REF_V              = 3.3      # Pico ADC reference
voltage

# Pump PWM frequency per Intel 4-wire spec
PUMP_PWM_FREQ_HZ      = 25000

```

C. Hardware Initialization

The pins to which each sensor and output is connected to must be defined clearly in the code. For example, the LCD is configured on GPIO 0 and GPIO 1. The thermocouple is created on GPIO 22, and the airflow sensor at GPIO 27. The pump PWM output is configured on GPIO 16 at 25 kHz with an initial duty of 0, ensuring the pump is off at startup. The pump tachometer input on GPIO 17 is configured as an input with a weak pull-up. Finally, the three warning indicators — the two LEDs and the buzzer are configured as outputs and initialized to off.

```

# I2C bus for OLED display (I2C0 on GPIO 0/1)
i2c = I2C(0, sda=Pin(0), scl=Pin(1), freq=400000)
oled = ssd1306.SSD1306_I2C(128, 64, i2c)

# 1-Wire bus for MAX31850K (GPIO 22)
ow_bus = onewire.OneWire(Pin(22))

# ADC for D6F-V03A1 air velocity sensor (GPIO 27 =
# ADC1)
flow_adc = ADC(Pin(27))

# Pump PWM output (GPIO 16) and tachometer input
# (GPIO 17)
pump_pwm = PWM(Pin(16))
pump_pwm.freq(PUMP_PWM_FREQ_HZ)
pump_pwm.duty_u16(0)
pump_tach = Pin(17, Pin.IN, Pin.PULL_UP)

# Warning indicators
temp_led = Pin(14, Pin.OUT)
flow_led = Pin(15, Pin.OUT)
buzzer = Pin(7, Pin.OUT)
temp_led.value(0)
flow_led.value(0)
buzzer.value(0)

```

D. MAX31850K Thermocouple Driver

Reading the MAX31850K requires a driver that works with the custom 1-Wire. The MicroPython defines a standard **ds18x20** module for reading DS18B20 temperature sensors, but it is not compatible with MAX31850K. The driver below should be included in the main code loop, and is written with

guidance from the datasheet obtained from Maxim Integrated [16].

```

# Fault flag bits from MAX31850K datasheet
MAX31850_FAULT_OC = 0x01      # Open circuit on
thermocouple
MAX31850_FAULT_SCG = 0x02     # Short to GND
MAX31850_FAULT_SCV = 0x04     # Short to VCC

def max31850_read_temp(ow):
    """Read thermocouple temperature and fault flags
    from MAX31850K."""
    try:
        # Issue Convert T command
        ow.reset()
        ow.writebyte(0xCC)     # Skip ROM (one
device on bus)
        ow.writebyte(0x44)     # Convert T
        time.sleep_ms(120)     # MAX31850K
conversion takes up to 100 ms

        # Read scratchpad
        ow.reset()
        ow.writebyte(0xCC)
        ow.writebyte(0xBE)     # Read scratchpad
        scratchpad = bytearray(9)
        for i in range(9):
            scratchpad[i] = ow.readbyte()

        # Extract 14-bit signed temperature in 0.25
deg C steps
        raw = (scratchpad[1] << 8) | scratchpad[0]
        fault_bit = raw & 0x0001
        temp_raw = raw >> 2
        if temp_raw & 0x2000:   # Sign extension
            temp_raw -= 0x4000
        temp_c = temp_raw * 0.25

        # Fault details in byte 2 lower 3 bits
        fault_flags = scratchpad[2] & 0x07

        if fault_bit:
            return (None, fault_flags)
        return (temp_c, 0)

    except Exception:
        return (None, 0xFF)     # 1-Wire bus
communication error

```

E. Airflow Sensor Reading

The D6F-V03A1 outputs an analog voltage that is proportional to air velocity, which is read by the firmware through the Pico's ADC. Linear interpolation is then used to convert this voltage reading into ms^{-1} between the calibrated zero-flow and full-flow voltages. It is suggested for the student to recalibrate the airflow when the system is operational to ensure no false readings.

During our tests, the output voltage stays below 3.3V at maximum flow. If this is found to increase beyond the Pico's logic level, the sensor should be connected to the pin with a voltage divider.

```

def read_airflow_ms():
    """Read air velocity from Omron D6F-V03A1 via
    ADC."""
    raw = flow_adc.read_u16()

```

```

voltage = (raw / 65535) * ADC_REF_V

if voltage <= FLOW_V_AT_ZERO:
    return 0.0
velocity = ((voltage - FLOW_V_AT_ZERO) /
            (FLOW_V_AT_FULL - FLOW_V_AT_ZERO)) *
FLOW_RANGE_MS
return max(0.0, velocity)

```

F. Cooling Control Logic

The banded PWM control law from Section V is implemented here as a single lookup function that maps temperature to duty cycle. A second function then applies the duty cycle to the pump's PWM output.

```

def compute_pump_duty(temp_c):
    """Map detector temperature to pump duty cycle
    (Table II)."""
    if temp_c < TEMP_BAND_OFF:
        return 0
    elif temp_c < TEMP_BAND_LOW:
        return 25
    elif temp_c < TEMP_BAND_MID:
        return 50
    elif temp_c < TEMP_BAND_HIGH:
        return 75
    else:
        return 100

def set_pump_duty(duty_percent):
    """Set pump PWM duty cycle (0-100 integer
    percent)."""
    duty_u16 = int((duty_percent / 100) * 65535)
    pump_pwm.duty_u16(duty_u16)

```

G. Warning System

The warning system activates the buzzer and indicator LED if any of the fault conditions is detected. Two different LEDs are used for the temperature or flow faults, while the same audible buzzer is shared to alert operators. A thermocouple fault is also added under the temperature faults as when this occurs, it cannot be guaranteed that the detector is operating within a safe range.

```

def update_warnings(temp_c, airflow_ms,
thermocouple_fault):
    """Set warning indicators based on current
    sensor readings."""
    temp_warn = (thermocouple_fault or
                 (temp_c is not None and temp_c >=
TEMP_WARNING))
    flow_warn = airflow_ms < FLOW_WARNING_MIN

    temp_led.value(1 if temp_warn else 0)
    flow_led.value(1 if flow_warn else 0)
    buzzer.value(1 if (temp_warn or flow_warn) else
0)

    return (temp_warn, flow_warn)

```

H. OLED Display

This display function refreshes the 128 x 64 OLED with the current system state at every loop iteration. The states of interest to us are the detector temperature, pump duty cycle, airflow velocity and warning status. The display layout takes

advantage of the LCD's two tone panel, where top 16 pixel rows are rendered in yellow for system header, while bottom 48 rows are rendered in blue for live data and status.

```

def update_display(temp_c, duty_percent, airflow_ms,
temp_warn, flow_warn, fault):
    """Refresh OLED with current sensor readings and
    warnings."""
    oled.fill(0)

    # Yellow header band (top 16 px of two-tone
    panel)
    oled.text("ME3 Cooling Sys", 0, 0)

    # Live readings in the blue body area
    if fault:
        oled.text("TEMP: FAULT", 0, 16)
    elif temp_c is not None:
        oled.text("TEMP: {:.1f} C".format(temp_c),
0, 16)
    else:
        oled.text("TEMP: ---", 0, 16)

    oled.text("PUMP: {:d}%".format(duty_percent), 0,
26)
    oled.text("FLOW: {:.2f} m/s".format(airflow_ms),
0, 36)

    # Status line at bottom
    status = ""
    if temp_warn:
        status += "!TEMP "
    if flow_warn:
        status += "!FLOW"
    if not status:
        status = "OK"
    oled.text("STATUS: " + status, 0, 52)

    oled.show()

```

I. Main Control Loop

The main loop assembles all the components from A-H above into a functioning flow. At each iteration, the sensors readings are taken, and the pump duty cycle is computed, with a 100% duty cycle if thermocouple fault occurs. On each loop, the warnings and display are also updated, with a CSV formatted log-line formatted to be used for test data collection.

```

def main():
    """Main control loop at LOOP_PERIOD_MS
    cadence."""
    print("ME3 Cooling Control System - starting")
    oled.fill(0)
    oled.text("ME3 Cooling Sys", 0, 0)
    oled.text("Initializing...", 0, 28)
    oled.show()
    time.sleep_ms(1000)

    while True:
        loop_start = time.ticks_ms()

        # Sensor reads
        temp_c, fault_flags =
max31850_read_temp(ow_bus)
        airflow_ms = read_airflow_ms()
        thermocouple_fault = (fault_flags != 0)

        # Cooling control with safe-state fallback
        on fault

```

```

if thermocouple_fault or temp_c is None:
    duty_percent = 100
else:
    duty_percent = compute_pump_duty(temp_c)
    set_pump_duty(duty_percent)

# Warnings and display
temp_warn, flow_warn = update_warnings(
    temp_c, airflow_ms, thermocouple_fault)
update_display(temp_c, duty_percent,
airflow_ms,
temp_warn, flow_warn,
thermocouple_fault)

# CSV serial log for bench testing
timestamp = time.time()
print("{}, {}, {}, {:.2f}, {}".format(
    timestamp,
    "{:.2f}".format(temp_c) if temp_c is not
None else "FAULT",
    duty_percent,
    airflow_ms,
    fault_flags))

# Maintain loop cadence
elapsed = time.time() - time.time()
loop_start)
sleep_ms = LOOP_PERIOD_MS - elapsed
if sleep_ms > 0:
    time.sleep(sleep_ms)

if __name__ == "__main__":
    try:
        main()
    except KeyboardInterrupt:
        # Clean shutdown for development use
        set_pump_duty(0)
        buzzer.value(0)
        temp_led.value(0)
        flow_led.value(0)
        print("Stopped by user")

```

VII. BENCH TESTING

A. Test Objective and Scope

The control system was bench-validated to confirm the function of the implemented stepped PWM scheme described in the previous section. The objective was to verify that the firmware correctly reads detector temperature from the thermocouple, that the pump duty cycle steps up according to temperature bands, and that OLED display updates in real time to reflect system state.

B. Test Setup

The control system was operated on the bench in its breadboard configuration, with the Pico powered via USB from the host computer. The pump and chiller are connected to their dedicated power supplies through the bidirectional shifter. The K-type thermocouple is secured with a standard handheld heat gun as the heat source. The heat gun is pointed at the thermocouple, allowing it to move through the full operating range of the control system.

The OLED display is used as the primary monitoring interface during the test, displaying the detector temperature,

pump duty cycle and air flow velocity. The buzzer and LEDs were also set to go off at the predetermined warning states.

C. Observed Behaviour

The tests were carried out with a different version of the firmware which had different thresholds for the duty cycles. Therefore, the results below do not align perfectly with the eventually decided upon thresholds. However, it still proves that the system works as designed, with the values shown in Figures 11 and 12. At temperatures below 35°C, the pump runs at a duty cycle of 50%. As the temperature steps up to above 35°C, the pump duty cycle changes to 100%, with an observable increase in water flow rate. The audible buzzer and LED warning light is also triggered to warn operators.

The tests showed that the display was able to refresh continuously throughout, with no observable lag between temperature change and the corresponding duty cycle update.

D. Future Tests

In future tests, the system should be verified against a fully powered ME3 detector setup, and run at extended amount of time to verify that the setup works as intended.

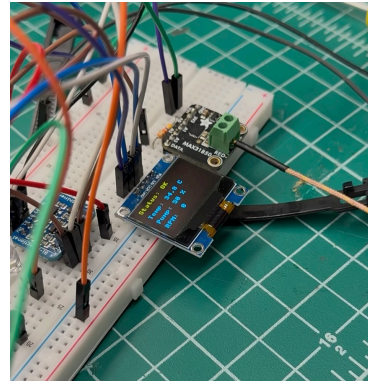


Fig. 11 Bench testing setup with the heatgun heating thermocouple to 50% duty cycle.

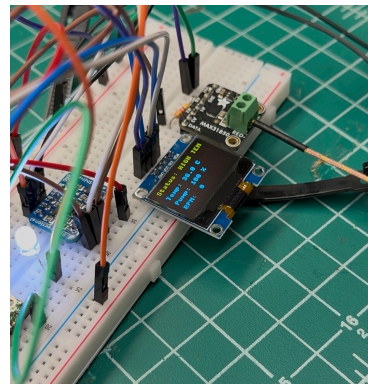


Fig. 12 Bench testing setup with the heatgun heating thermocouple to 100% duty cycle, triggering indicator LED and buzzer.

VIII. PCB TRANSFER

The breadboard prototype documented in the previous sections is then moved onto a KiCad project to support migration to a fabricated printed circuit board (PCB). PCB transfer is necessary especially for deployment on the rotating stage, where vibration, repeated handling and cable management constraints may make breadboard implementation impractical. Having soldered contacts on the PCB also eliminates the possibility of sensor faults under thermal cycling.

In this section, the paper discusses the schematic capture, footprint assignments and routed PCB layout in KiCad. All of this will form the basis for physical fabrication and assembly to be used in future work.

A. Schematic Capture

The complete electrical schematic is drawn in KiCad and shown in Figure 13 below. The schematic mirrors the breadboard exactly, with the Raspberry Pi Pico 2W as the ventral component, and the LCD display, thermocouple, bidirectional shifter, airflow sensor, buzzer and indicator LEDs connected as follows:

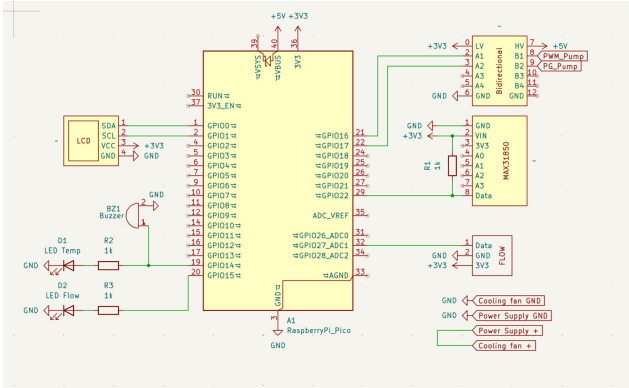


Fig. 13 Schematic drawing on KiCad for the translation of breadboard assembly into fully functioning PCB.

B. Footprint Assignment and Component Placement

Each component is then assigned with a PCB footprint. The Pico2W uses its standard 40 pin style header footprint. The MAX31850K and BSS138 use their breakout-board footprints with through hole headers, which allow them to be socketed instead of soldered, allowing for replacement during development. The OLED display uses a 4-pin header footprint that matches the GreekPi module's pin layout. The buzzer, LEDs and resistors use through-hole footprints measured with the parts in hand.

Component placement is mainly driven by two goals, which was to position the OLED in a clear front-facing position, and also to cluster the indicators (buzzer and LEDs) near the edge of the board for accessibility. The 3D render in Figure 14 is a representation of how the fabricated PCB would look like once manufactured.

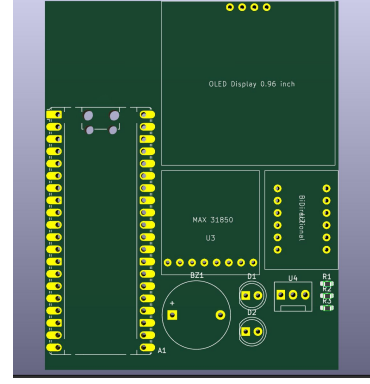


Fig. 14 KiCad 3D render of the unrouted PCB showing the placement of components on the board.

C. Routed Layout

The PCB is routed as a two-layer board with the routed traces as shown in Figure 15. The routing strategy implements a continuous ground return on the back layer to act as common ground for all components.

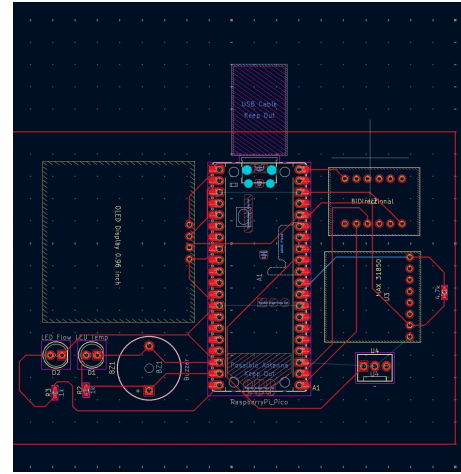


Fig. 15 Routed PCB layout with traces.

D. Outstanding Work for PCB Fabrication

Several items remain before the PCB can be integrated into the system, which include but are not limited to those listed below:

- **Fuse and protection placement**

The current breadboard implementation has no in-line fuses. The PCB layout should add footprints for the addition of fast blow fuses on each power input, which are the Pico's 5V input, pump 12V rail and chiller 12V rail.

- **Connector selection**

Each off-board connection like the pump cables, chiller cables, thermocouple wires and flow sensor wires should terminate at a labelled connector on the PCB edge. Examples

of connectors in this case would be screw terminals or JST connectors.

- **Fabrication and assembly**

Once the final layout and footprints are finalized, the PCB can be ordered from a standard prototyping service like JLCPCB. The student should then power the board up incrementally with current limits to verify that each rail is working as is, before the firmware is installed on the printed board.

IX. FUTURE WORK

The work done throughout this academic year developed a working, bench-validated cooling and dry air control system for the ME3 detector enclosure. This section will discuss some remaining items to be addressed before the system can be deployed on the rotating stage. This will allow for a clean handoff into the next development cycle. PCB fabrication is not mentioned here as the entirety of Section VIII documented the process in detail.

A. Migration to Closed-Loop PID Control

The most substantial upgrade that can be made is the migration from the current stepped PWM scheme to a closed-loop controller. This is already supported by the hardware in place, so the change made is purely to the software. The MAX31850K can provide temperature feedback at sub-second update rates, the pump can operate at continuous duty cycles between 0% and 100% while the Pico has enough processing power for the control computation.

It is personally recommended for the firmware to follow a single-loop PI controller architecture rather than full PID. The reason for this is that the thermal system has a slow time constant and significant deadtime. This negates the need for the derivative term, as it only amplifies thermocouple measurement noise far out of proportion.

Three main steps are implemented for PI control:

(1) Bench characterization of system thermal response

With the pump initially off, a known heat input is applied and the temperature is logged. The pump is then switched on at a fixed duty cycle and the temperature logged until steady state is achieved. The resulting temperature against time curve will produce three main parameters, which are **process gain** (K), **time constant** (τ) and **deadtime** (L).

(2) Initial gain selection

This can be found using the Cohen-Coon formula for systems with deadtime [17]:

$$K_p = \frac{0.9\tau}{KL} \text{ and } K_i = \frac{K_p}{3.33L}$$

(3) Manual refinement

This is done by observing the closed-loop response to a heat step disturbance and adjusting the gains iteratively until the best results are obtained.

The firmware can then be updated by making an update to the banded lookup in `compute_pump_duty` to instead have a PI computation that maintains state across loop iterations. The PI controller will calculate its output as the sum of a proportional term ($K_p \times \text{current error}$) and an integral term ($K_i \times \text{integrated error}$). It will then clamp the result to the [0,100] duty cycle range and write it to the pump like before. A failure mode of integral windup must also be implemented to prevent the error from accumulating even when the pump is saturated at 100%. A hardware timer should also be implemented to prevent inconsistent sample intervals that may introduce gain drift, ensuring uniform sample timing regardless of how long any loop iteration takes.

B. On-Detector Integration and Validation

The coolant assembly should be mounted with the thermocouple on an operating detector array to validate the system under realistic conditions.

The complete six detector array should be powered and run continuously for several hours to see if the cooling system can hold its temperature at the operating range. During this run, coolant temperature should be measured at several points along the loop to ensure it remains above the 15°C dew point margin.

Future cycles may choose to add a humidity sensor in the enclosure and hence make a change to the dry air subsystem. It can then be adapted from an always-on operation to a closed-loop humidity feedback control.

C. Standalone Operation and Deployment

Deployment also requires power supply to be clearly defined so it can operate on the rotating stage. Instead of using power supplies, the power must be wired to 12V wall plug adapters that can easily be powered with an extension cable. The Pico should also be connected by using a 12V → 5V buck converter so it can share the same power supply. This consolidates the systems into a few AC connection points, simplifying the wiring on the stage.

X. CONCLUSION

This report has documented the implementation and bench validation of the cooling and dry-air subsystem for the ME3 X-ray photon detector enclosure. Three main goals were achieved throughout the research process.

Firstly, the cascaded PID control architecture originally proposed by Cheng in [1] was reevaluated against the actual operational requirements. Since it is just a one-sided operating threshold and not a precise setpoint, it is replaced with a temperature banded open-loop PWM scheme. Secondly, both mechanical subsystems for cooling and dry air were assembled with a detailed bill of materials on the components

purchased for future reference. Specific instructions on the fitting and waterproofing were recorded to assist in any future maintenance efforts. Finally, the complete sensing and control electronics were prototyped on a breadboard, with firmware in MicroPython running the subsystems in parallel. Live status of the operation is rendered on an OLED display and audible and indicator warnings are triggered in the event of a fault.

All engineering decisions made in this project were documented here and driven primarily to match system complexity with actual operating requirements. The reason for doing so is that future cycles can easily revisit any decision should requirements change during operation of the detectors.

The complete electrical design was then captured as a fabrication ready KiCad schematic and routed PCB layout. Layout considerations and fabrication tasks have been listed out for PCB migration in the near future. Some suggestions are made in terms of the remaining outstanding work before the system can be fully deployed on the rotating stage.

ACKNOWLEDGMENT

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The author would also like to thank Jonathon Cheng, whose preceding work in [1] provided much of the foundation for the mechanical assembly and development of the control firmware.

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